

Adversarial audit of the kernel–Loewner floor

PRIME.RDAGGER.KERNEL_LOEWNER_REDTEAM.01 — round 495

Stefan Hamann

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Abstract

This round tries to kill the r494 assertion $\lambda_*(0.3) \geq 2.1 \cdot 10^{-3}$ from first principles. The identity was rederived from the defining digamma multiplier and cross-checked on five new compactly supported functions at 50-plus decimal precision. Translation boundary strips were checked exactly over $\mathbb{Q}$. A false doubled- c_L world is detected as negative. An independent Clenshaw–Curtis Nyström route reproduces the 401-section eigenvalue and Hilbert–Schmidt tail, including a nonzero complementary block that could not be discarded. No attack closes. The outward r494 balance is

$$3.69800804029 - 2.19240491114 - 1.5013673873 \\ - 3(7.0457565325 \cdot 10^{-4}) = 2.12201489024986 \cdot 10^{-3}.$$

Verdict: CONFIRMED($c = 2.1000\text{e-}3$).

Claim boundary. This is an adversarial audit of one unconditional fixed-support Weil inequality in the classical $2L < \log 2$ zone. It is not an RH claim, not an anti-RH claim, and creates no ledger row.

1 A1: independent derivation of the identity

Use

$$\widehat{h}(t) = \int_{\mathbb{R}} h(u) e^{itu} du, \quad g(x) = \int_{\mathbb{R}} h(u) h(u-x) du.$$

Then $(2\pi)^{-1} \int |\widehat{h}(t)|^2 \cos(tx) dt = g(x)$. Starting from the Gauss representation, independently of the r494 note,

$$\psi(z) = -\gamma + \int_0^\infty \frac{e^{-v} - e^{-zv}}{1 - e^{-v}} dv, \quad \operatorname{Re} z > 0,$$

and putting $z = \frac{1}{4} + \frac{i}{2}t$, $v = 2x$, gives

$$\sigma_A(t) = -\gamma - \log \pi + \int_0^\infty \frac{2e^{-2x} - 2e^{-x/2} \cos(tx)}{1 - e^{-2x}} dx.$$

The cosine coefficient is

$$k(x) = \frac{2e^{-x/2}}{1 - e^{-2x}} = \frac{e^{x/2}}{\sinh x}.$$

Plancherel therefore yields

$$A(h) = \int_0^\infty \left(\frac{e^{-2x}}{x} g(0) - k(x) g(x) \right) dx - (\log \pi) g(0).$$

For $\operatorname{supp} h \subset [-L, L]$, $g(x) = 0$ for $x > 2L$. The regularized elementary identity

$$\lim_{\varepsilon \downarrow 0} \left(\int_\varepsilon^{2L} \frac{dx}{x} - \int_\varepsilon^\infty \frac{e^{-2x}}{x} dx \right) = \log(4L) + \gamma$$

then gives

$$A(h) = \int_0^{2L} k(x)(g(0) - g(x)) dx - c_L g(0),$$

$$c_L = \int_0^{2L} \left(k(x) - \frac{1}{x} \right) dx + \log(4L) + \gamma + \log \pi = 2.1924049111319992552529 \dots$$

The pole sign is also forced rather than assumed. Analytic continuation gives

$$\widehat{g}(i/2) = \widehat{h}(i/2)\widehat{h}(-i/2) = (C - S)(C + S) = C^2 - S^2,$$

where $C = \langle h, \cosh(\cdot/2) \rangle$ and $S = \langle h, \sinh(\cdot/2) \rangle$. Thus

$$\Pi(h) = 2(C^2 - S^2).$$

It is a product of analytic values, not $|\widehat{h}(i/2)|^2$; the negative odd sign in r494 is correct. Since $e^{2L} < 2$ at $L = 0.3$, there is no prime term.

Five non-calibration cross-tests

Set $u = x/L$ and $h_j(x) = (1 - u^2)^8 p_j(u)$ on $[-L, L]$, zero outside, with

$$p_1 = 1, \quad p_2 = u, \quad p_3 = 1 + \frac{7}{10}u - \frac{2}{5}u^2,$$

$$p_4 = 1 - 2u^2 + \frac{3}{10}u^3, \quad p_5 = 1 - \frac{1}{3}u + \frac{1}{5}u^2 - \frac{1}{7}u^3.$$

These are not the r494 calibration modes. The frequency side uses $\operatorname{Re} \psi(\frac{1}{4} + it/2) - \log \pi$ directly at 60 working digits; the x side integrates the independently derived kernel.

j	$Q_{\text{W}}^{\text{freq}}/\ h_j\ ^2$	$Q_{\text{W}}^x/\ h_j\ ^2$	normalized difference
1	0.26745365580067245	0.26745365580067245	$-1.873 \cdot 10^{-22}$
2	0.85826315683789947	0.85826315683789947	$-5.736 \cdot 10^{-21}$
3	0.28721078110996577	0.28721078110996577	$-1.506 \cdot 10^{-22}$
4	0.33249049031538647	0.33249049031538647	$-2.305 \cdot 10^{-22}$
5	0.26385733080320823	0.26385733080320823	$-3.014 \cdot 10^{-22}$

The c_L computations at 50/80-digit settings differ by $2.12 \cdot 10^{-61}$ in the full run.

2 A2: exact boundary-strip attack

For the zero extension E and $(\tau_x f)(y) = f(y + x)$,

$$\langle Eh, \tau_x Eh \rangle = g(x), \quad \|Eh - \tau_x Eh\|_{L^2(\mathbb{R})}^2 = 2(g(0) - g(x)).$$

Two rational step functions were used. The first has values $(2, -1, 3)$ on the three cells cut at $-3/10, -1/10, 1/10, 3/10$. The second has values $(-2, 5/2)$ on the two half intervals. Exact interval intersection arithmetic gives:

step	x	$2(g_0 - g_x)$	whole-line norm	interval-only norm	omitted strip
1	1/20	19/10	19/10	17/10	1/5
1	1/10	19/5	19/5	17/5	2/5
1	1/4	13/2	13/2	113/20	17/20
2	1/20	61/40	61/40	53/40	1/5
2	1/10	61/20	61/20	53/20	2/5
2	1/4	61/8	61/8	53/8	1

All six whole-line identities are exact. Every interval-only norm is strictly too small. R494 uses the whole-line norm and retains the two boundary strips.

3 A3: the false-world test

Replace c_L by $2c_L$ while leaving k, w, Π fixed. This control functional is known negative: the h_1 test above has Rayleigh quotient

$$0.26745365580067245 - c_L = -1.9249512553313268.$$

The independent truncated-kernel chain also becomes

$$\kappa_w - 2c_L - \lambda_{\max}(T_w - \Pi) = -2.188167590810956.$$

Thus the machinery does not prove positivity in the negative control world. It fails by an order-one margin, as required.

4 A4: independent numbers

The alternative route uses Clenshaw–Curtis quadrature on the physical square $[-L, L]^2$ and a symmetric Nyström matrix. It does not use the r494 Gauss–Legendre/Chebyshev surrogate to obtain its diagnostic numbers.

$$\kappa_w = 3.69800804029897744651553 \dots$$

CCorder	$\lambda_{\max}(S)$	$\lambda_{\max}(P_{401}SP_{401})$	r_{401}	$\ QTP\ _{\text{HS}}$	$\ QTQ\ _{\text{HS}}$
513	1.501362199357172	1.501362199357172	$5.952545 \cdot 10^{-4}$	$1.740576 \cdot 10^{-4}$	$5.419740 \cdot 10^{-4}$
769	1.501365757748623	1.501365757748621	$6.826870 \cdot 10^{-4}$	$1.816422 \cdot 10^{-4}$	$6.325138 \cdot 10^{-4}$
1025	1.501365808845935	1.501365808845941	$6.794869 \cdot 10^{-4}$	$1.804971 \cdot 10^{-4}$	$6.297174 \cdot 10^{-4}$

At order 1025 the sampled Legendre Gram defect is $6.335 \cdot 10^{-12}$. The r494 section centre 1.501365606961153 differs by $2.02 \cdot 10^{-7}$ and is safely under the outward upper bound 1.5013673873. The independent tail lies below the certified r494 upper bound $7.0457565325 \cdot 10^{-4}$.

An adaptive square-root endpoint transform gives the uncut $N = 36$ even diagnostic $7.571961013293050 \cdot 10^{-3}$. Combining it with the independent cut section gives a diagnostic cutoff loss $3.334640692256198 \cdot 10^{-3}$, within $2.02 \cdot 10^{-7}$ of the r494 value. The load-bearing direct interval subtraction is

$$\begin{aligned} & (3.69800804029 - 2.19240491114 - 1.5013673873) \\ & - 3(7.0457565325 \cdot 10^{-4}) = 2.122014890249863 \cdot 10^{-3} > 2.1 \cdot 10^{-3}. \end{aligned}$$

5 A5: the full operator-class check

Let $P = P_{401}$ and $Q = I - P$. Because $w = 0$ near zero and the quintic ramp matches through two derivatives, $w(x)k(x)$ is C^2 on $[0, 2L]$. Hence

$$T_w(y, z) = \frac{1}{2}w(|y - z|)k(|y - z|)$$

is bounded on a compact square and is Hilbert–Schmidt. Smoothness supplies the observed rate; Hilbert–Schmidt membership alone supplies convergence.

Normalized Legendre polynomials are a complete orthonormal basis of $L^2[-L, L]$. Parseval in the Hilbert–Schmidt class therefore gives the exact identity

$$r_n^2 = \|T_w\|_{\text{HS}}^2 - \sum_{i,j < n} |(T_w)_{ij}|^2 = 2\|QT_wP\|_{\text{HS}}^2 + \|QT_wQ\|_{\text{HS}}^2.$$

Thus the old off-space problem cannot hide in an unrepresented class: the sum contains *both* off-blocks and the entire block-diagonal rest beyond n . Since operator norm is at most Hilbert–Schmidt norm,

$$2\|PT_wQ\| + \|QT_wQ\| \leq 3r_n.$$

The independent order-1025 decomposition finds the block-diagonal rest explicitly:

$$\|QT_wP\|_{\text{HS}} = 1.8049712 \cdot 10^{-4}, \quad \|QT_wQ\|_{\text{HS}} = 6.2971739 \cdot 10^{-4}.$$

It is not small enough to discard, but it is included. Directly,

$$2\|QT_wP\|_{\text{HS}} + \|QT_wQ\|_{\text{HS}} = 9.9071162 \cdot 10^{-4} < 3(7.0457565325 \cdot 10^{-4}).$$

The rank-two pole vectors are entire. On $|x/2| \leq 0.15$, the degree-400 Taylor remainder is bounded by $e^{0.15} 0.15^{401}/401!$, far below 10^{-30} even after the rank-two operator conversion. No pole remainder is hidden.

6 Kill table and verdict

Attack	Result
A1	Survives: kernel, c_L , $g(0)$ term, pole sign, and five direct frequency/ x comparisons agree.
A2	Survives: the zero-extension identity is exact; both boundary-strip masses are retained.
A3	Survives: a known negative doubled- c_L world makes both the Rayleigh quotient and the certificate budget negative.
A4	Survives: independent quadrature reproduces κ_w , section eigenvalue, cutoff loss, tail, and final arithmetic inside all outward bounds.
A5	Survives: completeness plus Hilbert–Schmidt Parseval includes both off-blocks and the nonzero QTQ remainder.

Theorem 1 (Red-team verdict). *The $r494$ assertion survives all five attacks:*

$$Q_W(h) \geq 2.1 \cdot 10^{-3} \|h\|_2^2 \quad \text{for } \text{supp } h \subset [-0.3, 0.3].$$

*The red-team verdict is **CONFIRMED** ($c = 2.1000\text{e-}3$).*

No RH claim and no anti-RH claim is made.